

BUCHANAN FIELD, CA, USA

WMO: 724950

Lat: 37.992N

Lon: 122.055W

Elev: 18

StdP: 14.69

Time Zone: -8.00 (NAP)

Period: 99-19

WBAN: 23254

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 33.1	(c) 35.5	(d) 18.7	(e) 14.1	(f) 54.8	(g) 22.9	(h) 17.3	(i) 54.2	(j) 19.6	(k) 55.5	(l) 17.2	(m) 53.5	(n) 1.9	(o) 160	(p) 0.491

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	28.6	97.9	66.9	94.2	65.8	90.6	64.9	69.0	92.8	67.5	89.9	66.2	87.1	10.1	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 60.3	(b) 78.3	(c) 73.6	(d) 59.0	(e) 74.6	(f) 71.8	(g) 57.9	(h) 71.5	(i) 70.5	(j) 33.0	(k) 92.7	(l) 31.8	(m) 89.9	(n) 30.7	(o) 87.1	(p) 76.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 19.8	(b) 18.1	(c) 16.6	(e) 29.8	(f) 104.8	(g) 2.1	(h) 3.1	(i) 28.3	(j) 107.0	(k) 27.1	(l) 108.8	(m) 26.0	(n) 110.5	(o) 24.4	(p) 112.7		
(a) 19.8	(b) 18.1	(c) 16.6	(e) 27.7	(f) 71.6	(g) 2.3	(h) 2.0	(i) 26.0	(j) 73.1	(k) 24.7	(l) 74.3	(m) 23.4	(n) 75.4	(o) 21.7	(p) 76.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	61.4	49.2	52.3	56.3	59.3	64.7	70.3	72.7	72.4	70.7	64.3	55.2	49.2		
	DBStd	10.00	4.44	4.59	5.00	5.45	5.45	5.88	5.15	4.47	5.28	5.04	5.23	4.86		
	HDD50	190	69	26	6	1	0	0	0	0	0	0	15	73		
	HDD65	2263	490	356	273	187	72	12	3	2	8	74	295	491		
	CDD50	4367	44	90	201	280	457	610	705	696	622	444	171	47		
	CDD65	967	0	0	3	16	64	173	243	233	180	54	1	0		
	CDH74	11188	0	7	74	297	918	2110	2664	2346	2012	725	35	0		
	CDH80	5178	0	0	6	87	366	1035	1354	1116	976	235	3	0		
(14) Wind	WSAvg	7.2	4.5	5.6	6.6	7.9	9.4	10.2	10.4	9.5	7.4	5.6	4.5	4.7		
(15) Precipitation	PrecAvg	20.1	4.3	3.6	2.6	1.4	0.7	0.2	0.0	0.0	0.1	1.1	2.3	4.0		
	PrecMax	30.5	10.5	12.9	8.3	4.6	3.0	1.9	0.1	0.8	0.6	4.3	6.8	10.2		
	PrecMin	8.7	0.2	0.0	0.0	0.2	0.0	0.0	0.0	0.0	0.0	0.1	0.0	0.0		
	PrecStd	5.3	2.9	2.6	2.3	1.1	0.8	0.4	0.0	0.2	0.1	1.1	1.7	2.9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	68.2	74.3	80.5	87.8	95.3	101.8	102.5	100.2	99.7	91.0	78.2	66.0		
		MCWB	51.9	56.8	61.1	63.7	66.1	68.4	68.5	67.6	66.3	62.9	58.8	53.9		
	2%	DB	63.6	69.1	75.8	82.0	88.9	96.1	96.8	95.2	95.1	86.2	73.5	63.0		
		MCWB	52.9	55.4	59.3	61.2	63.7	66.5	67.0	66.3	64.9	61.2	57.3	53.2		
	5%	DB	60.9	65.1	71.7	76.6	83.7	91.0	92.2	90.7	90.6	81.9	69.3	60.5		
		MCWB	52.2	53.8	57.1	59.0	61.6	64.9	66.2	65.2	64.0	59.4	55.6	53.0		
	10%	DB	58.1	61.9	67.6	71.7	78.7	85.9	88.0	86.3	85.6	77.6	65.7	58.3		
		MCWB	51.3	53.0	55.4	57.1	60.0	63.0	65.1	64.4	62.7	58.6	54.7	52.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	58.0	60.3	63.4	65.7	68.7	70.8	71.9	69.7	69.5	65.9	62.3	59.5		
		MCDB	61.8	68.1	76.0	83.1	90.7	97.3	98.4	93.6	95.4	83.9	72.9	62.7		
	2%	WB	55.9	58.1	60.7	62.5	65.4	68.0	68.8	68.0	67.1	63.7	60.2	57.4		
		MCDB	60.0	64.8	72.9	79.0	86.4	92.0	92.5	90.4	88.6	80.3	68.6	60.0		
	5%	WB	54.1	56.2	58.6	60.2	62.9	66.0	67.2	66.4	65.4	62.1	58.6	55.4		
		MCDB	57.8	62.7	68.7	74.5	80.5	88.1	89.3	87.2	85.2	76.8	65.9	58.3		
	10%	WB	52.4	54.3	56.6	57.9	60.8	64.1	65.6	65.1	63.9	60.3	56.9	53.4		
		MCDB	56.4	60.4	65.2	69.8	76.5	83.8	86.3	84.3	82.7	72.9	63.6	56.4		
(35) Mean Daily Temperature Range	MDBR		16.5	17.9	19.6	21.1	23.5	26.8	28.6	27.5	27.1	24.2	19.8	16.3		
	5% DB	MCDBR	21.8	24.6	27.2	30.6	33.3	36.1	36.0	35.2	36.2	32.2	25.5	19.7		
		MCWBR	12.7	13.4	13.5	14.0	12.9	12.3	11.9	11.4	11.9	12.3	12.3	11.5		
	5% WB	MCDBR	15.0	19.0	23.3	27.6	30.3	33.3	34.3	31.7	31.3	26.8	20.1	11.8		
		MCWBR	10.2	11.4	12.6	13.4	12.8	12.3	12.1	11.1	11.1	11.3	11.1	8.5		
(40) Clear-Sky Solar Irradiance	taub		0.323	0.326	0.340	0.359	0.364	0.361	0.366	0.350	0.348	0.333	0.325	0.326		
	taud		2.536	2.529	2.475	2.394	2.407	2.421	2.406	2.444	2.446	2.525	2.530	2.529		
	Ebn at Noon		273	288	293	292	291	290	288	291	286	281	270	263		
	Edh at Noon		24	28	32	37	37	37	37	35	33	27	24	23		
(44) All-Sky Solar Radiation	RadAvg		736	1045	1481	1945	2293	2532	2454	2192	1821	1316	883	649		
	RadStd		114	130	142	161	141	104	74	76	64	85	58	100		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (47 neighbors)	+0.67	N/A	-2.07	N/A	N/A	N/A	N/A	N/A	-207	+239	N/A

Nomenclature: See separate page